Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/27/2007
PI Name:	Guo, Boyun Ph.D.		
Project Title:	Development of a CFD Model for Predicting Performance of Volatile Removal Assembly in the Closed Environment Life Support System under Microgravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	70504-4690	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05GA35G		
Performance Goal No.:			
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Task Description:	Volatile Removal Assembly (VRA) is one of the key components of the Closed Environment Life Support System (CELSS) currently used in the International Space Station. Its purpose is to remove volatile organics to convert the wastewater into reusable water in the space station. It is believed that the performance of the VRA depends on several key factors including properties of catalyst pack and fluids, flow rates of water and oxygen, pressure, temperature, geometry of the VRA, oxidation kinetics, and level of gravitational force. However, it is not fully understood how these factors control the efficiency of the VRA. The objective of the proposed research project is to develop and validate a Computational Fluid Dynamics (CFD) model to predict the performance of the VRA under microgravity conditions. Such a CFD model can be used for designing high performance VRA's. The method of approach to achieving this objective is to combine the efforts of mathematical modeling, computer simulation, laboratory investigations, and VRA performance measurements from KC-135 ballistic trajectory flight. The mathematical modeling will include numerical formulation of fluid dynamics for computing fluid-phase distribution in the VRA with considerations of all the above-mentioned factors. The computer simulation will include coding the developed numerical formulation in a modern computer to create a fluid dynamics simulator. The laboratory investigations will include measurements of the absolute and relative permeabilities of a catalyst pack to water and oxygen under different conditions. They will also include measurements of oxidation kinetics. These investigations will determine values of parameters to be used in the computer model. NASA has made the VRA-performance measurements from a KC-135 ballistic trajectory flight under certain conditions. The results will be used in this research project to validate the CFD model.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	NASA has successfully designed, tested and implemented in the International Space Station a Volatile Removal Assembly (VRA) as part of a Closed Environment Life Support System (CELSS). The VRA contains 4 catalyst-packed reactors performing catalyst oxidation of water at elevated pressure and temperature under microgravity conditions. Although the technology has been proven to be successful, the current designing practice relies on an empirical method in engineering calculations. It has been unclear as to how much deviation from the current design is allowed in the future VRA designs. The results of this project is a validated CFD model that can be used for designing high performance VRA's.
Task Progress:	To achieve the goal of developing a Computational Fluid Dynamics (CFD) model for designing high performance Volatile Removal Assembly (VRA) used in the Closed Environment Life Support System (CELSS), we conducted research in two areas: 1) experimental studies of water-nitrogen two-phase flow in sand packs under chemical equilibrium conditions, and 2) numerical simulation of two-phase flow in porous media under non-equilibrium conditions. Tests were run at different combinations of water and nitrogen flow rates. Nitrogen gas holdup values were determined at various inclination angles to account for the effect of gravity on two-phase flow. Nitrogen gas distributions were determined with X-Ray CT Scanning. A CFD model was developed to simulate two-phase reaction-flow in porous media. The following conclusions are drawn: 1) The end-affected length is from 2 cm to 5 cm in a 4.45 cm inner diameter tube. 2) Gravity segregation effect in all the horizontal flow experiments was not significant. Therefore, the flow conditions in the horizontal experiments are representative to those in the VRA reactor under microgravity environment. 3) Gas holdup in the sand packs increases as gravity drops. 4) Gas holdup increases with both water and gas injection rates. 5) A CFD model can be used to compute gas holdup value with acceptable error. Future work of this study will focus on conducting laboratory testing of reactor performance with wastewater and oxygen gas under non-equilibrium conditions to determine reaction rate constant and gas bubble size. The measured data and VRA performance measurements from KC-135 ballistic trajectory flight will be used for calibrating the developed CFD model.
Bibliography Type:	Description: (Last Updated: 10/12/2006)